

MADISON COOTS

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RESEARCH FOCUS

Responsible AI, inequality, discrimination, and the use of computational methods to advance decision-making and policy across a variety of domains, including in healthcare, lending, criminal justice, and education.

EDUCATION

Harvard Kennedy School <i>Ph.D. in Public Policy</i>	Cambridge, MA August 2022 - Present
Harvard University <i>M.A. in Public Policy</i>	Cambridge, MA August 2022 - May 2025
Stanford University <i>M.S. in Computer Science</i> <i>Specialization: Artificial Intelligence</i>	Stanford, CA September 2019 - June 2021
Stanford University <i>Bachelor of Science - Management Science and Engineering</i> <i>Minor: English</i>	Stanford, CA September 2015 - June 2019

PEER-REVIEWED PUBLICATIONS

- [1] Alex Chohlas-Wood, [Madison Coots](#), Joe Nudell, Julian Nyarko, Emma Brunskill, Todd Rogers, and Sharad Goel. “Automated Court Date Reminders Reduce Warrants for Arrest: Evidence from a Text Messaging Experiment.” *Science Advances*. 2025.
- [2] [Madison Coots](#), Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. “Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.” *Annual Review of Public Health*. 2025.
- [3] [Madison Coots](#), Soroush Saghaian, David Kent, and Sharad Goel. “A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk.” *Annals of Internal Medicine*. 2024.
- [4] Alex Chohlas-Wood, [Madison Coots](#), Henry Zhu, Sharad Goel, and Emma Brunskill. “Learning to be Fair: A Consequentialist Approach to Equitable Decision-Making.” *Management Science*. 2024.
- [5] Alex Chohlas-Wood*, [Madison Coots](#)*, Julian Nyarko*, and Sharad Goel*. “Designing Equitable Algorithms.” *Nature Computational Science*, Vol. 3. 2023. *Authors contributed equally.

BOOK CHAPTERS

- [1] [Madison Coots](#), Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. “Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.” Chapter in *Social Stratification, 5th Edition* (2027). Edited by Michelle Jackson, Nima Dahir, Claire Daviss, and David B Grusky. Routledge.

RESEARCH IN PROGRESS

- [1] [Madison Coots](#), Robert Bartlett, Julian Nyarko, and Sharad Goel. “Algorithmic Bias in Lending: Evidence from a Fintech Audit.” *Under review*. 2026.
- [2] [Madison Coots](#), Hans Gaebler, Josh Grossman, Alex Chohlas-Wood, Julian Nyarko, and Sharad Goel. “Towards Individualized Risk Predictions with LLMs.” *In preparation*. 2026.
- [3] [Madison Coots](#), Calvin Isley, Elisa Xi Chen, Nick Masi, Joe Nudell, Charlotte Tuminelli, Teddy Svoronos, and Sharad Goel. “AI Tutoring and Teacher Support Tools for Elementary Education.” *In preparation*. 2026.

HONORS, AWARDS, AND FELLOWSHIPS

Distinction in Student Teaching Award • Harvard Kennedy School	2026
Dissertation Research Grant, \$10,000 • Horowitz Foundation for Social Policy	2026
Rising Star in IOE-ISyE-MS&E • Workshop jointly hosted by University of Michigan, Georgia Tech, and Stanford University	2026
Dean's Award for Excellence in Student Teaching • Harvard Kennedy School	2025
Distinction in Student Teaching Award • Harvard Kennedy School	2025
Social Equity and Health Equity Grant, \$5,000 • Malcolm Weiner Center for Social Policy, Harvard Kennedy School	2025
James M. and Cathleen D. Stone PhD Scholar in Inequality and Wealth Concentration • Stone Program in Wealth Distribution, Inequality, and Social Policy, Harvard University	2023 - Present
Harvard Graduate Prize Fellowship • Harvard University	2022 - 2023
Stanford Engineering Coterminial Fellowship • Stanford University	2019 - 2020
Stokes Scholar • U.S. Federal Government	2017 - 2021

SELECTED PRESENTATIONS

International Conference on Computational Social Science <i>"Algorithmic Bias in Lending: Evidence from a Fintech Audit"</i> (Submitted talk)	Burlington, VT 2026
Econometric Society ESIF: Economics and AI+ML Conference <i>"Algorithmic Bias in Lending: Evidence from a Fintech Audit"</i> (Submitted talk)	Ithaca, NY 2026
Production and Operations Management Society Annual Conference <i>"A Profit-Based Measure of Lending Discrimination"</i> (Submitted talk)	Reno, NV 2026
IOE-ISyE-MS&E Rising Stars Workshop <i>"A Profit-Based Measure of Lending Discrimination"</i> (Invited talk and poster)	Ann Arbor, MI 2026
American Law and Economics Association Conference <i>"A Profit-Based Measure of Lending Discrimination"</i> (Submitted talk)	New York, NY 2025
Production and Operations Management Society Annual Conference <i>"A Profit-Based Measure of Lending Discrimination"</i> (Invited talk)	Atlanta, GA 2025
Predictive Analytics & Comparative Effectiveness Center Symposium <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	Boston, MA 2024
Society of Medical Decision Making 46th Annual Meeting <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Submitted talk)	Boston, MA 2024
International Conference on Computational Social Science <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Poster)	Philadelphia, PA 2024
Computational and Methodological Statistics Conference <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	Remote 2023
APPAM Fall Research Conference <i>"Automated Court Date Reminders Reduce Warrants for Arrest"</i> (Submitted talk)	Atlanta, GA 2023
INFORMS General Meeting <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	Phoenix, AZ 2023
INFORMS Healthcare Conference <i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	Toronto, ON 2023
ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization <i>"Learning to be Fair"</i> (Submitted talk with Alex Chohlas-Wood)	Arlington, VA 2022

TEACHING

Harvard Kennedy School

- **Teaching Fellow.** API 201: Quantitative Analysis and Empirical Methods I (Fall 2024, 2025). *Graduate course in applied statistics: exploring and summarizing data with R, probability theory, statistical inference, decision analysis.*
- **Teaching Fellow.** API 202Z: Quantitative Analysis and Empirical Methods II (Spring 2025, 2026). *Advanced graduate course in applied statistics: statistical inference and linear regression models.*
- **Teaching Fellow.** API 203Z: Quantitative Analysis and Empirical Methods III (Spring 2025). *Advanced graduate course in applied statistics: causal inference and machine learning.*
- **Teaching Fellow.** DPI 681M: The Science and Implications of Generative AI (Spring 2024) *Designed and taught the technical complement to the course on language models, deep learning models, and transformers.*

Stanford University, School of Engineering

- **Course Assistant.** MS&E 125: Applied Statistics (Winter 2020) *Undergraduate course in applied statistics: exploring and summarizing data, statistical inference, linear and logistic regression models.*
- **Course Assistant.** MS&E 252: Foundations of Decision Analysis (Fall 2019). *Graduate course in quantitative decision analysis covering: utility theory, sensitivity analysis, value of information. Recognized by Stanford Center for Professional Development for excellence in teaching.*

Stanford Code in Place Program

- **Section Leader.** (Spring 2020). *Part of a teaching team for Code in Place, offered by Stanford during COVID-19 pandemic, with 10,000 global students and 900 volunteer teachers participating from around the world. Prepared and taught a weekly discussion section of 10-12 students to supplement professors' lectures in a 5-week introductory online Python programming course.*

PROFESSIONAL SERVICE

Ad Hoc Peer Reviewer

- *Epidemiology* (2025)
- *International Conference on Computational Social Science* (2025, 2026)

Organizing Team Member

- *Teaching Statistics in the Age of Agentic AI*, Harvard Kennedy School (June 2026)

Supporting Instructor

- *An Introduction to Reproducible Programming and Project Management*, Society of Medical Decision Making Annual Meeting (2024)

SKILLS SUMMARY

Languages: R, Python

AI: LLM prompt and response evaluation, text data processing and information extraction, chatbot design and user experience for AI systems

Methods: Stochastic modeling, decision analysis, statistical inference, causal inference

PROFESSIONAL EXPERIENCE

Senior Data Scientist (Part-time)

Systems & Technology Research

Woburn, MA

Feb. 2023 - Present

Data Scientist

Stanford Computational Policy Lab

Boston, MA

Sep. 2020 - Aug. 2022

Data Scientist (Part-time)

Aerospace Technical Services

Remote

Sep. 2020 - Jun. 2024

Data Science Intern

U.S. Federal Government (agency withheld)

Washington D.C.

Jun. 2017 - Jan. 2021